



Indian Institute of Technology Guwahati
Department of Electronics and Electrical Engineering
EE-210 (Digital Circuits)
Assignment Problem
Set: 44

1. Design and implement a fully functional synchronous First-In, First-Out (FIFO) buffer in Verilog. Your design should be generic and configurable using parameters for data width and FIFO depth.

Inputs: `CLOCK`, `RESET`, `WR_EN` (Write enable signal. Data is written when this signal is high and the FIFO is not full), `RD_EN` (Read enable signal. Data is read when this signal is high and the FIFO is not empty), and `WR_DATA` (The data to be written into the FIFO).

Outputs: `RD_DATA` (The data read from the FIFO), `FULL` (A flag that is high when the FIFO is at its maximum capacity), and `EMPTY` (A flag that is high when the FIFO contains no data).

The module should be parameterized as follows:

- `DATA_WIDTH`: The bit width of the data.
- `DEPTH_BITS`: The number of bits for the FIFO pointers.

Your design must meet the following functional requirements:

- **Synchronous Operation:** All state changes (pointer increments, data writes) must occur on the rising edge of the `CLOCK` signal.
- **Synchronous Reset:** The `RESET` signal should reset the FIFO to a completely empty state, clearing both the read and write pointers.
- **Read/Write Logic:**
 1. A write operation should only occur when `WR_EN` is high and the full flag is low.
 2. A read operation should only occur when `RD_EN` is high and the empty flag is low.
 3. If both `WR_EN` and `RD_EN` are high in the same clock cycle, the data should be written and a new data item should be read. The total count of items in the FIFO should remain unchanged.
- **FIFO Pointers and Flags:** Design the logic for the full and empty flags. Explain how your design avoids the ambiguity of using a single pointer comparison (where `WR_PTR == RD_PTR` could mean both full and empty).

Specify the FPGA target used for implementation. Synthesize and implement the design using Verilog HDL, then report the LUTs, flip-flops, delay, maximum operating frequency, and power for each.

2. Implement a Verilog module for the following 4-bit adder architectures::

- Carry select adder
- Carry bypass adder

Specify the FPGA target used for implementation. Synthesize and implement the designs using Verilog HDL, then report the following:

- Propagation Delay
- Max operating frequency
- Hardware area